STAT 4C03

```
data = read.csv("Fastener.csv")
head(data)
```

```
##   Pressure Num.Fasteners Failed
## 1     2500            50     10
## 2     2700            70     17
## 3     2900           100     30
## 4     3100            60     21
## 5     3300            40     18
## 6     3500            85     43
```

a)

```
model.a = glm(
  cbind(Failed, Num.Fasteners - Failed) ~ Pressure,
  data = data,
  family = binomial
)
summ.a = summary(model.a)
summ.a
```

```
##
## Call:
## glm(formula = cbind(Failed, Num.Fasteners - Failed) ~ Pressure,
##     family = binomial, data = data)
##
## Coefficients:
##               Estimate Std. Error z value Pr(>|z|)
## (Intercept) -5.3397115  0.5456932  -9.785   <2e-16 ***
## Pressure     0.0015484  0.0001575   9.829   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 112.83207  on 9  degrees of freedom
## Residual deviance:   0.37192  on 8  degrees of freedom
## AIC: 49.088
##
## Number of Fisher Scoring iterations: 3
```

Analysis result:

The results indicate that pressure has a significant positive effect on the probability of fastener failure. As pressure increases, the odds of failure increase, suggesting that higher pressure loads are associated with a greater risk of fastener failure.

b)

```
# Satured model
model.b = glm(
  cbind(Failed, Num.Fasteners - Failed) ~ factor(Pressure),
  data = data,
  family = binomial
)
summ.b = summary(model.b)
summ.b
```

```
##
## Call:
## glm(formula = cbind(Failed, Num.Fasteners - Failed) ~ factor(Pressure),
##     family = binomial, data = data)
##
## Coefficients:
##                      Estimate Std. Error z value Pr(>|z|)
## (Intercept)           -1.3863     0.3536  -3.921 8.82e-05 ***
## factor(Pressure)2700   0.2492     0.4502   0.554 0.579886
## factor(Pressure)2900   0.5390     0.4155   1.297 0.194527
## factor(Pressure)3100   0.7673     0.4453   1.723 0.084862 .
## factor(Pressure)3300   1.1856     0.4754   2.494 0.012634 *
## factor(Pressure)3500   1.4098     0.4148   3.399 0.000677 ***
## factor(Pressure)3700   1.7918     0.4139   4.329 1.50e-05 ***
## factor(Pressure)3900   2.0496     0.4627   4.429 9.46e-06 ***
## factor(Pressure)4100   2.4849     0.4378   5.676 1.38e-08 ***
## factor(Pressure)4300   2.6791     0.4648   5.764 8.22e-09 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance:  1.1283e+02  on 9  degrees of freedom
## Residual deviance: -1.9984e-15  on 0  degrees of freedom
## AIC: 64.717
##
## Number of Fisher Scoring iterations: 3
```

```
# fitted failure probabilities from both models
fitted.table <- data.frame(
  Pressure = data$Pressure,
  Saturated = predict(model.b, type = "response"),
  Logistic  = predict(model.a, type = "response")
)

fitted.table
```

```
##    Pressure Saturated  Logistic
## 1      2500 0.2000000 0.1871513
## 2      2700 0.2428571 0.2388598
## 3      2900 0.3000000 0.2995894
```

```
## 4      3100 0.3500000 0.3682883
## 5      3300 0.4500000 0.4427816
## 6      3500 0.5058824 0.5199410
## 7      3700 0.6000000 0.5961606
## 8      3900 0.6600000 0.6680059
## 9      4100 0.7500000 0.7327982
## 10     4300 0.7846154 0.7889409
```

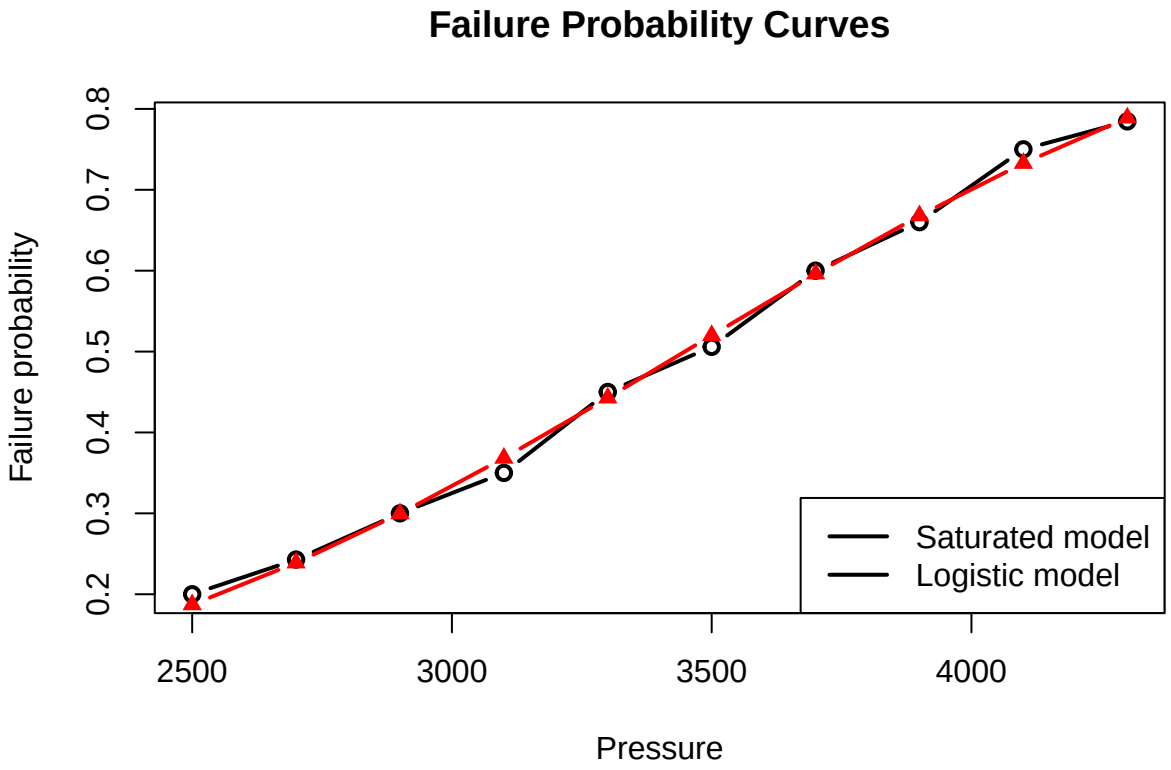
c)

```r
# Extract variables
x  <- fitted.table$Pressure
pS <- fitted.table$Saturated
pL <- fitted.table$Logistic

# Plot the saturated model curve
plot(x, pS,
     type = "b",      # both points and lines
     lwd = 2,
     xlab = "Pressure",
     ylab = "Failure probability",
     main = "Failure Probability Curves")

# Add the logistic model curve
lines(x, pL,
      type = "b",
      col="red",
      pch = 17,
      lwd = 2)

# Add legend
legend("bottomright",
       legend = c("Saturated model", "Logistic model"),
       lwd = 2)
```



The two curves are almost exactly the same with slight differences.

d)

```
# Computing the deviance
ni = data$Num.Fasteners
pi.hat.i = model.a$fitted.values
yi = data$Failed

D = 2 * sum( yi * log( yi / (ni * pi.hat.i) ) + (ni - yi) * log( (ni - yi) / (ni * (1 - pi.hat.i)) ) )
cat("Computed deviance =", D, "model.a's deviance =", model.a$deviance)
```

```
## Computed deviance = 0.3719169 model.a's deviance = 0.3719169
```

We have exactly the same value of the deviance, the results agree.

e)

```
# Pearson GOF statistic

X2 <- sum((yi - ni*pi.hat.i)^2 / (ni*pi.hat.i*(1 - pi.hat.i)))

n  <- length(yi) # number of observations
p  <- 1 # estimated parameters
```

```r
df <- n - p - 1 # degree of freedom

p_value <- 1 - pchisq(X2, df)

cat("Pearson X2 =", X2, "\n")
```

```
## Pearson X2 = 0.3706631
```

```r
cat("p-value =", p_value, "\n")
```

```
## p-value = 0.9999576
```

```r
# The deviance GOF p-value
p_value = 1 - pchisq(D, df)
cat("p-value =", p_value, "\n")
```

```
## p-value = 0.999957
```

The Pearson and deviance goodness-of-fit tests both yield large p-values around 0.99. There is no evidence against the fitted binomial regression model. The model appears to provide an adequate fit to the data.

f)

```r
# The model with no covariate
model.f = glm(
  cbind(Failed, Num.Fasteners - Failed) ~ 1,
  data = data,
  family = binomial
)

# LRT
anova(model.a, model.f)
```

```
## Analysis of Deviance Table
##
## Model 1: cbind(Failed, Num.Fasteners - Failed) ~ Pressure
## Model 2: cbind(Failed, Num.Fasteners - Failed) ~ 1
##   Resid. Df Resid. Dev Df Deviance  Pr(>Chi)
## 1         8      0.372
## 2         9    112.832 -1  -112.46 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The value of the test sttatistic is G2 =112.83207 − 0.37192= 112.46015 Under H0 (Pressure has no effect of the the probability of failure), G2 follows chi2(1) The corresponding p-value is p-value = 2.2e-16 ## Conclusion Since the p-value is extremely small, we reject the null hypothesis. There is an overwhelming evidence that pressure significantly affects the probability of failure.